

A HYBRID CAUSAL MACHINE LEARNING FRAMEWORK FOR MARKETING SPEND OPTIMIZATION

Submitted in Partial Fulfilment of the Requirements for the Award of

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By

Mr. SIDHARTH V

A869117724003

Under the Guidance of

Dr. KRISHNACHALITHA K C

Assistant Professor-AIIT



AMITY INSTITUTE OF INFORMATION TECHNOLOGY

AMITY UNIVERSITY, BENGALURU

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Name: SIDHARTH V

SEN: A869117724003

ABSTRACT

Marketing organizations invest significant resources across multiple advertising channels to improve customer acquisition and maximize revenue. However, traditional predictive machine learning models primarily identify correlations between marketing expenditure and business outcomes, making it difficult to determine the true causal impact of individual marketing channels. This limitation often leads to suboptimal budget allocation decisions.

This dissertation proposes a Hybrid Causal Machine Learning Framework for Marketing Spend Optimization, which integrates predictive machine learning with causal inference. A marketing dataset containing advertising expenditures, customer engagement metrics, and purchase information was preprocessed and enhanced through feature engineering to generate meaningful variables such as Total Marketing Spend, Click-Through Rate (CTR), Return on Advertising Spend (ROAS), and Log Revenue.

Two machine learning models, Random Forest Regressor and XG Boost Regressor, were developed and evaluated for revenue prediction. The Random Forest model achieved superior predictive performance with an R^2 Score of 0.8640 and a Mean Absolute Error (MAE) of 0.3596, making it the preferred predictive model for subsequent analysis. To estimate the causal contribution of individual marketing channels, the study employed the Causal Forest Double Machine Learning (Causal Forest DML) approach. The estimated Average Treatment Effects (ATEs) identified Google Shopping, Meta Facebook, and Google Paid Search as the most effective marketing channels in terms of their positive causal impact on revenue.

Furthermore, a counterfactual simulation engine was developed to evaluate alternative marketing budget allocation strategies. By simulating different budget reallocation scenarios, the framework demonstrated that shifting investment toward channels with higher positive causal effects can improve predicted revenue. Among the evaluated scenarios, reallocating budget from TikTok to Google Shopping resulted in the highest predicted revenue improvement of approximately 0.46% over the baseline strategy.

The proposed framework demonstrates that combining predictive analytics with causal inference provides deeper business insights than prediction alone. By identifying both revenue forecasts and the true effectiveness of marketing channels, the framework serves as a practical decision-support tool for optimizing marketing budgets and improving return on investment.

Keywords: Marketing Spend Optimization, Causal Machine Learning, Causal Forest DML, Random Forest, XG Boost, Counterfactual Simulation, Marketing Analytics, Budget Allocation, Predictive Analytics.

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1.INTRODUCTION

1.1 Motivation

In today's highly competitive digital marketplace, organizations invest substantial portions of their budgets across multiple marketing channels such as Google Ads, Meta (Facebook and Instagram), TikTok, and other online advertising platforms. Although these investments generate vast amounts of customer engagement data, determining the actual contribution of each marketing channel to business outcomes remains a significant challenge. Traditional marketing analytics primarily rely on predictive models that estimate future sales based on historical patterns, but they often fail to distinguish between correlation and true causation. As a result, marketing managers may allocate budgets inefficiently, leading to suboptimal return on investment (ROI) and reduced campaign effectiveness.

Recent advancements in machine learning and causal inference provide an opportunity to address these limitations by combining predictive accuracy with cause-and-effect analysis. Machine learning algorithms can effectively model complex and nonlinear relationships between marketing investments and revenue, while causal inference techniques enable the estimation of the genuine impact of individual marketing channels by accounting for confounding factors. Furthermore, counterfactual simulation allows organizations to evaluate alternative budget allocation strategies before implementing them in real-world campaigns. Motivated by these advancements, this research proposes a hybrid causal machine learning framework that integrates predictive modelling, causal effect estimation, and counterfactual simulation to optimize marketing spend across multiple channels and support data-driven strategic decision-making.

1.2 Issues and Challenges

Optimizing marketing spend across multiple advertising channels has become increasingly challenging due to the growing complexity of digital marketing ecosystems. Organizations invest simultaneously in platforms such as Google Ads, Meta (Facebook and Instagram), TikTok, and other digital channels, making it difficult to accurately measure the individual contribution of each channel to overall business performance. Some of the major challenges associated with marketing spend optimization include:

- Complex and nonlinear relationships between marketing investments and revenue, making traditional statistical models less effective.
- Interactions among multiple marketing channels, where the performance of one channel may influence or depend on the spending in another channel.
- High-dimensional and heterogeneous marketing data, consisting of varying spend levels, clicks, impressions, and customer engagement metrics collected from multiple sources.
- Difficulty in distinguishing correlation from causation, as conventional predictive models identify patterns but cannot determine whether a specific marketing channel genuinely causes an increase in revenue.
- Dynamic market conditions and changing consumer behaviour, which continuously affect campaign performance and reduce the effectiveness of static budget allocation strategies.

Traditional Marketing Mix Models (MMM) and regression-based approaches often rely on simplifying assumptions and may not adequately capture nonlinear relationships or accurately estimate the causal impact of marketing activities. Although advanced machine learning models provide improved predictive performance, they generally function as black-box models and lack the capability to explain the true cause-and-effect relationships between marketing investments and business outcomes. These limitations highlight the need for a hybrid framework that combines machine learning with causal inference techniques to support more reliable marketing budget optimization and data-driven decision-making.

1.3 Problem Definition

Organizations invest significant amounts of their marketing budgets across multiple digital advertising channels such as Google Paid Search, Google Shopping, Meta (Facebook and Instagram), TikTok, and other online platforms. Despite the availability of large volumes of campaign data, determining the true effectiveness of each marketing channel remains a challenging task. Most existing marketing analytics techniques focus on predicting business outcomes using historical data but are unable to distinguish between correlation and causation. Consequently, marketing managers often make budget allocation decisions based on predictive insights that may not accurately reflect the actual impact of individual marketing channels.

This research addresses the problem by developing a hybrid causal machine learning framework that combines predictive modelling with causal inference techniques. The proposed framework aims to accurately estimate the causal contribution of individual marketing channels to revenue generation while maintaining high predictive performance. Furthermore, a counterfactual simulation engine is incorporated to evaluate alternative budget allocation scenarios and identify strategies that maximize return on investment (ROI). By integrating machine learning, causal effect estimation, and simulation-based optimization, the proposed system provides a reliable decision-support framework for optimizing multi-channel marketing spend.

1.4 Objectives

The primary objectives of this study are:

- To develop a hybrid causal machine learning framework for optimizing marketing spend across multiple digital advertising channels.
- To preprocess and engineer marketing campaign data by generating meaningful features such as Total Marketing Spend, Click-Through Rate (CTR), Return on Ad Spend (ROAS), and Log Revenue.
- To develop and compare the performance of Random Forest Regressor and XG Boost Regressor for predicting marketing revenue using multi-channel advertising data.
- To evaluate the predictive models using performance metrics such as R^2 Score and Mean Absolute Error (MAE) and identify the most suitable model for revenue prediction.
- To estimate the causal impact of individual marketing channels using the Causal Forest Double Machine Learning (DML) approach and quantify the Average Treatment Effect (ATE) of each marketing channel.
- To design a counterfactual simulation engine for evaluating different marketing budget allocation scenarios and predicting their impact on revenue.

- To recommend an optimized marketing budget allocation strategy that maximizes revenue and improves Return on Investment (ROI) through data-driven decision making

1.5 Report Organization

This dissertation is organized into six chapters, each focusing on a specific aspect of the research work.

- Chapter 1 – Introduction: Presents the background of the study, motivation, problem definition, research objectives, issues and challenges, and the overall organization of the dissertation.
- Chapter 2 – Literature Review: Reviews existing research on Marketing Mix Modelling (MMM), multi-channel marketing analytics, machine learning techniques, causal inference, and counterfactual analysis. It also identifies the research gap that motivates the proposed framework.
- Chapter 3 – Proposed Methodology: Describes the complete research methodology, including dataset description, data preprocessing, feature engineering, predictive model development using Random Forest and XG Boost, causal effect estimation using Causal Forest DML, and the counterfactual simulation framework for marketing spend optimization.
- Chapter 4 – Implementation: Explains the implementation of the proposed system, including the software environment, Python libraries used, module-wise implementation, model training, causal analysis, and simulation of budget allocation scenarios.
- Chapter 5 – Results and Discussion: Presents the experimental results obtained from the predictive models and causal inference analysis. It evaluates model performance using R^2 Score and MAE, discusses the estimated Average Treatment Effects (ATE) of different marketing channels, and analyses the outcomes of counterfactual marketing budget simulations.
- Chapter 6 – Conclusion and Future Work: Summarizes the major findings of the research, highlights the contributions of the proposed hybrid causal machine learning framework, discusses its limitations, and suggests possible directions for future research and practical enhancements.
- References: Lists all the books, research papers, journals, and online resources cited throughout the dissertation in the prescribed citation format.

2.LITERATURE SURVEY

2.1 Marketing Mix Modelling (MMM)

Marketing Mix Modelling (MMM) is a widely adopted analytical approach used to evaluate the effectiveness of various marketing activities on business performance. It enables organizations to quantify the contribution of different marketing channels, such as search advertising, social media campaigns, television, email marketing, and other promotional activities, towards key business outcomes including sales, revenue, customer acquisition, and return on investment (ROI). By examining historical marketing and sales data, Marketing Mix Modelling provides valuable insights that assist organizations in making informed budget allocation decisions and improving overall marketing efficiency.

Traditional Marketing Mix Modelling primarily relies on statistical techniques such as Multiple Linear Regression to establish relationships between marketing expenditure and business outcomes. These models assume that each marketing channel contributes independently to sales and that the relationship between marketing investment and revenue is linear. Although these assumptions simplify the modelling process and improve interpretability, they often fail to capture the complex interactions among modern digital marketing channels. As consumer behaviour becomes increasingly dynamic and marketing campaigns span multiple online platforms, traditional regression-based models struggle to represent nonlinear relationships and cross-channel dependencies accurately.

The rapid expansion of digital advertising platforms such as Google Ads, Meta (Facebook and Instagram), TikTok, and YouTube has significantly increased the complexity of marketing data. Organizations now generate large volumes of high-dimensional campaign data containing information related to advertising spend, impressions, clicks, conversions, customer engagement, and purchasing behaviour. This evolution has exposed several limitations of conventional Marketing Mix Models, including their inability to model nonlinear relationships, limited scalability with large datasets, and difficulty in adapting to changing market conditions.

To overcome these challenges, researchers have increasingly explored the integration of advanced machine learning techniques within Marketing Mix Modelling. Machine learning algorithms are capable of identifying hidden patterns, capturing nonlinear relationships, and processing large-scale marketing datasets without requiring strict statistical assumptions. These capabilities have improved the accuracy of marketing performance prediction while enabling businesses to gain deeper insights into customer behaviour and campaign effectiveness. Consequently, modern Marketing Mix Modelling has gradually shifted from purely statistical approaches towards data-driven predictive analytics supported by artificial intelligence and machine learning methodologies.

Despite these advancements, conventional Marketing Mix Modelling primarily focuses on measuring associations between marketing expenditure and business outcomes rather than identifying true causal relationships. As a result, organizations may allocate marketing budgets based on observed correlations that do not necessarily represent the actual contribution of individual marketing channels. This limitation has encouraged researchers to integrate causal inference techniques with Marketing Mix Modelling to better estimate the

genuine impact of marketing investments and support more effective budget optimization strategies.

2.2 Machine Learning in Marketing Analytics

Machine learning has emerged as one of the most influential technologies in modern marketing analytics due to its ability to analyse large volumes of structured and unstructured data while identifying complex patterns that are difficult to detect using traditional statistical methods. Organizations increasingly employ machine learning algorithms to improve customer segmentation, demand forecasting, recommendation systems, customer lifetime value prediction, campaign optimization, and sales forecasting. By learning from historical marketing data, machine learning models can generate accurate predictions that support strategic business decision-making and enhance overall marketing performance.

Unlike conventional regression techniques, machine learning algorithms are capable of modelling highly nonlinear relationships between marketing variables and business outcomes. Digital marketing environments involve numerous interacting factors such as advertising expenditure, customer engagement, seasonal variations, promotional campaigns, and market competition. These factors often exhibit nonlinear dependencies that cannot be adequately represented using traditional statistical models. Machine learning algorithms overcome these limitations by automatically identifying complex interactions within the data, thereby improving predictive accuracy.

Among various machine learning algorithms, ensemble learning techniques have gained significant attention due to their robustness and superior predictive performance. Random Forest is an ensemble-based algorithm that constructs multiple decision trees and combines their predictions to reduce overfitting while improving generalization. Its ability to handle high-dimensional datasets, missing values, and nonlinear relationships makes it highly suitable for marketing analytics applications. Random Forest has been successfully applied in customer purchase prediction, campaign performance evaluation, churn prediction, and revenue forecasting.

Another widely adopted ensemble learning algorithm is Extreme Gradient Boosting (XG Boost), which employs gradient boosting techniques to sequentially improve prediction accuracy by minimizing residual errors. XG Boost has demonstrated excellent performance across numerous predictive analytics applications due to its computational efficiency, regularization mechanisms, and ability to model complex feature interactions. In marketing analytics, XG Boost has been extensively utilized for sales prediction, advertising effectiveness analysis, customer response modelling, and campaign optimization.

Although machine learning models provide highly accurate predictions, they generally function as black-box models that explain what is likely to happen rather than why it happens. Their predictions are primarily based on observed data patterns and correlations, making it difficult to determine whether a particular marketing activity genuinely causes changes in business outcomes. Consequently, researchers have increasingly focused on integrating causal inference techniques with machine learning models to obtain both accurate predictions and meaningful explanations for marketing decision-making.

2.3 Causal Inference in Marketing

Causal inference is a statistical framework that aims to estimate the true cause-and-effect relationships between variables rather than merely identifying correlations. In marketing analytics, understanding causal relationships is essential because organizations require reliable evidence regarding how changes in advertising expenditure influence sales, revenue, or customer acquisition. While predictive machine learning models accurately estimate future business outcomes, they do not necessarily explain whether a particular marketing channel is responsible for those outcomes. This distinction between prediction and causation has become a significant research focus in recent years.

One of the primary concepts in causal inference is the Average Treatment Effect (ATE), which measures the average impact of a treatment variable on an outcome variable. In the context of marketing analytics, the treatment variable typically represents marketing expenditure on a specific advertising channel, while the outcome variable represents business performance measures such as revenue or sales. Estimating ATE enables organizations to quantify the actual contribution of individual marketing channels after accounting for other influencing factors.

Recent developments in machine learning have led to advanced causal inference techniques such as Double Machine Learning (DML) and Causal Forests. These methods combine machine learning algorithms with statistical estimation procedures to reduce bias caused by confounding variables while accurately estimating heterogeneous treatment effects. Causal Forest DML extends traditional random forest models by incorporating orthogonalization techniques that separate prediction from causal estimation, thereby improving the reliability of treatment effect estimation in observational datasets.

The application of causal inference in marketing has enabled businesses to move beyond correlation-based analytics towards evidence-based decision-making. By estimating the true causal impact of individual marketing channels, organizations can allocate advertising budgets more effectively, improve return on investment, and design marketing strategies based on measurable business impact rather than historical associations.

3.SYSTEM SPECIFICATION

3.1 Software Requirements

The proposed Hybrid Causal Machine Learning framework was developed using Python and its associated data science libraries. The implementation environment was selected to facilitate efficient data preprocessing, predictive modelling, causal inference, visualization, and marketing spend optimization.

Operating System: Windows 11

Programming Language: Python 3.10+

Development Environment / IDE:

- Visual Studio Code (VS Code)
- Jupyter Notebook

Version Control:

- Git
- GitHub

Python Libraries and Frameworks:

- Pandas – Data loading, cleaning, and manipulation
- NumPy – Numerical computations and array operations
- Matplotlib – Data visualization
- Seaborn – Statistical data visualization
- Scikit-learn – Machine learning algorithms and model evaluation
- XG Boost – Gradient boosting regression model
- Econ ML – Causal inference using Causal Forest Double Machine Learning (DML)

3.2 Language and Technology

The proposed Hybrid Causal Machine Learning framework was developed using Python and a collection of advanced data science, machine learning, and causal inference libraries. The selected technologies facilitate efficient data preprocessing, predictive modelling, causal effect estimation, visualization, and marketing spend optimization.

Programming Language

- Python 3.10+

Python was selected due to its extensive ecosystem of libraries for data analytics, machine learning, causal inference, and scientific computing. Its flexibility and ease of integration make it suitable for developing scalable marketing analytics applications.

Technologies Used

1. Data Preprocessing and Feature Engineering
 - Data cleaning and missing value handling
 - Feature creation using Pandas and NumPy
 - Generation of derived marketing metrics such as:
 - Total Marketing Spend
 - Total Clicks
 - Total Impressions
 - Click-Through Rate (CTR)
 - Return on Ad Spend (ROAS)
 - Log Revenue
2. Machine Learning Models
 - Random Forest Regressor
 - XG Boost Regressor

These models were employed to predict marketing revenue based on multi-channel advertising expenditure and customer engagement metrics.

3. Causal Inference
 - Causal Forest Double Machine Learning (DML) using the Econ ML library
The causal model estimates the Average Treatment Effect (ATE) of each marketing channel, enabling the identification of channels that genuinely influence revenue generation.
4. Counterfactual Simulation
 - Scenario-based budget reallocation
 - Revenue prediction under alternative marketing investment strategies
 - Marketing spend optimization
5. Data Visualization
 - Matplotlib
 - Seaborn

These libraries were used to visualize data distributions, feature relationships, channel rankings, and model performance.

Methodology

The proposed system follows a structured Hybrid Causal Machine Learning methodology consisting of the following stages:

1. Data Collection and Integration
2. Data Cleaning and Preprocessing
3. Feature Engineering
4. Predictive Model Development
5. Model Evaluation
6. Causal Effect Estimation using Causal Forest DML
7. Counterfactual Budget Simulation
8. Marketing Spend Optimization
9. Result Visualization and Analysis

3.3 Hardware Requirements

The minimum and recommended hardware specifications required to implement the proposed Hybrid Causal Machine Learning framework are as follows:

Minimum Hardware Requirements

- **Processor:** Intel Core i5 (8th Generation) or AMD Ryzen 5
- **RAM:** Minimum 8 GB
- **Storage:** Minimum 100 GB free disk space
- **Graphics:** Card: Integrated Graphics
- **Others:** Stable Internet connection, Monitor with 1080p resolution

Recommended Hardware Requirements

- **Processor:** Intel Core i7 (10th Generation or above) or AMD Ryzen 7
- **RAM:** Minimum 16 GB
- **Storage:** 256 GB SSD or higher
- **Graphics:** Card: NVIDIA GeForce GTX 1650 or higher (Optional)
- **Others:** High-speed Internet connection, Monitor with 1080p or higher resolution

4.SYSTEM DESIGN

4.1 Methodology

This study proposes a Hybrid Causal Machine Learning framework for optimizing marketing budget allocation across multiple advertising channels. The methodology combines predictive machine learning models with causal inference techniques to estimate the true impact of marketing investments on revenue and recommend optimal budget allocation strategies. The overall methodology consists of the following stages:

• Data Collection

A large-scale multi-channel marketing dataset containing 132,759 records was collected. The dataset includes daily marketing expenditure, clicks, impressions, and purchase-related information across multiple digital advertising platforms, including:

- Google Paid Search
- Google Shopping
- Google PMAx
- Meta Facebook
- Meta Instagram
- TikTok

The dataset also contains purchase revenue and customer engagement metrics used for predictive and causal analysis.

• Data Preprocessing

The collected dataset was cleaned and prepared for analysis through the following steps:

- Handling missing values
- Removing duplicate records
- Converting data types
- Formatting date attributes
- Creating derived features such as:
 - Total Marketing Spend
 - Total Clicks
 - Total Impressions
 - Click-Through Rate (CTR)
 - Return on Ad Spend (ROAS)
 - Log Revenue
 - Year, Month, Quarter, and Day of Week

• Predictive Model Development

Machine learning models were developed to predict marketing revenue based on advertising expenditure and customer engagement features.

The following predictive models were implemented:

1. **Random Forest Regressor** – To capture complex nonlinear relationships between marketing channels and revenue.
2. **XG Boost Regressor** – To improve predictive accuracy using gradient boosting techniques.

The dataset was divided into 80% training data and 20% testing data for model development and evaluation.

• Causal Effect Estimation

To identify the true causal impact of individual marketing channels on revenue, the study employed the Causal Forest Double Machine Learning (DML) algorithm using the Econ ML framework.

For each marketing channel:

- **Treatment Variable:** Marketing spent of the selected channel
- **Outcome Variable:** Log Revenue
- **Control Variables:** Remaining marketing channels, clicks, impressions, and temporal features

The model estimates the Average Treatment Effect (ATE) for each advertising channel, enabling the identification of channels that have the greatest causal influence on revenue.

• Counterfactual Simulation

A counterfactual simulation engine was developed to evaluate different marketing budget allocation strategies. Multiple investment scenarios were created by increasing or decreasing advertising expenditure across selected marketing channels. The trained Random Forest model was then used to predict the expected revenue for each scenario.

• Marketing Spend Optimization

The predicted revenues obtained from different budget allocation scenarios were compared with the baseline revenue. The scenario producing the highest revenue improvement was identified as the optimal marketing budget allocation strategy.

• Evaluation Metrics

The predictive performance of the machine learning models was evaluated using the following metrics:

- Coefficient of Determination (R^2 Score)
- Mean Absolute Error (MAE)

The effectiveness of the causal model was evaluated using:

- Average Treatment Effect (ATE)

The counterfactual simulations were assessed by comparing the percentage improvement in predicted revenue under different marketing budget allocation strategies.

• **Tools Used**

The implementation was carried out using Python with the following libraries:

- Pandas
- NumPy
- Scikit-learn
- XG Boost
- Econ ML
- Matplotlib
- Seaborn

5. IMPLEMENTATION

5.1 Environment Setup

To develop, implement, and evaluate the proposed Hybrid Causal Machine Learning framework, a suitable software and hardware environment was established. The development environment consists of the following components:

1. Operating System

- Windows 11 (64-bit)

2. Programming Language

- Python 3.10

3. Development Environment / IDE

- Visual Studio Code (VS Code)
- Jupyter Notebook

4. Key Python Libraries and Packages

Data Processing & Analysis

- Pandas
- NumPy

Data Visualization

- Matplotlib
- Seaborn

Machine Learning

- Scikit-learn (Random Forest Regressor, Linear Regression for baseline comparison, model evaluation)

Gradient Boosting

- XG Boost (XG Boost Regressor)

Causal Inference

- Econ ML (Causal Forest Double Machine Learning)

Model Evaluation

- Scikit-learn Metrics
 - R^2 Score
 - Mean Absolute Error (MAE)

5. Version Control

- Git
- GitHub

6. Hardware Configuration

- **Processor:** Intel Core i5 (10th Generation) or equivalent
- **RAM:** 16 GB
- **Storage:** Minimum 100 GB free disk space
- **Graphics:** Integrated Graphics / NVIDIA GPU (Optional)
- **Display:** Monitor with 1080p resolution
- **Internet:** Stable internet connection for package installation and repository management

7. Additional Tools

- Git for version control
- GitHub for source code repository and project management

5.2.1 Module 1: Data Collection Module

Objective

To collect and organize multi-channel marketing data required for predictive modeling and causal analysis.

Description

This module is responsible for importing the marketing dataset containing advertising expenditure, customer engagement metrics, and purchase information. The collected dataset consists of **132,759 records** spanning multiple digital marketing channels.

The dataset includes:

- Google Paid Search Spend
- Google Shopping Spend
- Google PMAx Spend
- Meta Facebook Spend
- Meta Instagram Spend

- TikTok Spend
- Clicks
- Impressions
- Purchase Revenue

Key Functions

- Import CSV dataset using Pandas.
- Verify data structure and attribute types.
- Handle missing values and duplicate records.
- Prepare the dataset for preprocessing.

5.2.2 Module 2: Data Preprocessing & Feature Engineering Module

Objective

To clean, transform, and prepare the dataset for predictive modeling and causal inference.

Description

This module improves data quality by handling missing values, formatting variables, and creating additional features that enhance model performance.

The following features were generated:

- Total Marketing Spend
- Total Clicks
- Total Impressions
- Click Through Rate (CTR)
- Return on Ad Spend (ROAS)
- Log Revenue
- Year
- Month
- Quarter
- Day of Week

Key Functions

- Missing value handling
- Data type conversion
- Feature engineering
- Date transformation
- Revenue transformation using logarithmic scaling

5.2.3 Module 3: Predictive Machine Learning Module

Objective

To develop machine learning models capable of accurately predicting marketing revenue.

Description

The processed dataset was divided into **80% training data** and **20% testing data**. Two machine learning algorithms were trained and evaluated for revenue prediction.

Models Implemented:

- Random Forest Regressor
- XG Boost Regressor

The Random Forest model achieved the highest prediction accuracy and was selected for subsequent counterfactual simulations.

Performance Metrics

- R^2 Score
- Mean Absolute Error (MAE)

Key Functions

- Train-test split
- Model training
- Revenue prediction
- Performance evaluation

5.2.4 Module 4: Causal Inference & Counterfactual Simulation Module

Objective

To estimate the causal contribution of each marketing channel and evaluate alternative budget allocation strategies.

Description

This module applies the **Causal Forest Double Machine Learning (DML)** algorithm to estimate the **Average Treatment Effect (ATE)** of individual marketing channels on revenue.

Each marketing channel is considered a treatment variable while controlling for the remaining channels and engagement metrics.

Based on the estimated causal effects, multiple counterfactual scenarios were created by reallocating marketing budgets across different channels. The trained Random Forest model was then used to predict revenue under each scenario.

Key Functions

- Average Treatment Effect (ATE) estimation
- Marketing channel ranking
- Budget reallocation
- Counterfactual revenue prediction
- Scenario comparison

5.2.5 Module 5: Marketing Spend Optimization & Visualization Module

Objective

To identify the optimal marketing budget allocation strategy and visualize analytical results for decision support.

Description

This module compares the predicted revenue from different counterfactual scenarios with the baseline revenue to determine the most effective budget allocation strategy.

The analytical results are presented through various visualizations to facilitate interpretation and support strategic marketing decisions.

Visualizations Generated

- Revenue Distribution
- Correlation Heatmap
- Feature Importance
- Marketing Channel Causal Effect Ranking
- Counterfactual Scenario Comparison
- Revenue Improvement Analysis

Tools Used

- Matplotlib
- Seaborn
- Pandas

5.3 Database Implementation

This project does not utilize any external Database Management System (DBMS) such as MySQL, PostgreSQL, MongoDB, or Firebase. Instead, a **file-based storage approach** was adopted for storing, processing, and managing the marketing dataset throughout the implementation.

Justification for Not Using a Database

The decision to use CSV-based storage was based on the following reasons:

- The marketing dataset containing **132,759 records** was of manageable size and could be processed efficiently using Python.
- The **Pandas** library provides powerful functions for reading, writing, filtering, and manipulating CSV files without requiring a dedicated database system.
- The project focuses on **machine learning, causal inference, and marketing spend optimization**, rather than developing a transactional web application.
- No real-time data streaming, concurrent user access, or complex relational queries were required.
- File-based storage simplified experimentation, preprocessing, feature engineering, and model development.

File-Based Storage Approach

The project follows a structured directory-based storage approach.

Input Data

- Raw multi-channel marketing dataset stored in **CSV format**.

Processed Data

- Cleaned and feature-engineered datasets stored as CSV files.

Model Outputs

- Trained model predictions.
- Causal effect estimation results.
- Counterfactual simulation results.
- Performance evaluation metrics.

Project Repository

- Source code, notebooks, and datasets were managed using **Git** and **GitHub** for version control and collaboration.

Tools Used

- **Pandas** – Data loading, preprocessing, and CSV file management.
- **NumPy** – Numerical computations.

- **OS & Pathlib** – File and directory handling.
- **Jupyter Notebook** – Data processing and experimentation.
- **Visual Studio Code** – Project development and implementation.
- **Git & GitHub** – Version control and project repository management.

Project Directory Structure

Hybrid-MMM-Model/

```

|
|— data/
|   |— raw/
|   |— processed/
|
|— notebooks/
|   |— 01_data_preprocessing.ipynb
|   |— 02_feature_engineering.ipynb
|   |— 03_model_training.ipynb
|   |— 04_causal_model.ipynb
|   |— 05_counterfactual_simulation.ipynb
|
|— results/
|   |— causal_effects.csv
|   |— model_metrics.csv
|   |— simulation_results.csv
|
|— figures/
|
|— README.md

```

5.4 Algorithm

The proposed Hybrid Causal Machine Learning framework combines predictive machine learning with causal inference to estimate the true impact of marketing channels on revenue and optimize marketing budget allocation. The algorithm follows the steps below:

Algorithm: Hybrid Causal Machine Learning for Marketing Spend Optimization

Input: Multi-channel marketing dataset containing marketing spend, clicks, impressions, and purchase revenue.

Output: Optimized marketing budget allocation with predicted revenue improvement

Step 1: Load the raw marketing dataset from CSV files.

Date	Organization Vertical	Territory	Google Paid Search Spend (USD)	Google PMAX Spend (USD)	Meta Facebook Spend (USD)	All Purchases	Purchase Revenue (USD)
2022-07-29	Beauty & Fitness	US	0.00	205.53	233.75	27	4552.98
2022-07-30	Beauty & Fitness	US	0.00	277.08	248.84	17	3177.99
2022-07-31	Beauty & Fitness	US	0.00	427.44	274.51	39	5468.99
2022-08-01	Beauty & Fitness	US	35.24	347.23	270.06	22	3335.97
2022-08-02	Beauty & Fitness	US	38.40	314.95	257.62	28	3991.99

Table 5.1: Sample of the raw multi-channel marketing dataset used in this study

("Due to space limitations, only a representative subset of the dataset is presented in Table 5.1. The complete dataset consists of 50 attributes and 132,759 records.")

The raw marketing dataset contains 132,759 observations and 50 attributes representing daily marketing activities across multiple advertising channels. The dataset includes marketing expenditure, customer engagement metrics (clicks and impressions), purchase information, and organizational details. These records form the foundation for data preprocessing, feature engineering, predictive modelling, causal inference, and counterfactual simulation performed in this study.

Step 2: Data Preprocessing

The raw marketing dataset was preprocessed to improve data quality and prepare it for feature engineering and machine learning. The preprocessing stage included handling missing values, removing duplicate records, converting data types, and formatting date attributes.

5.2.1 Handling Missing Values

Missing values were identified using the `isnull()` function in Pandas. Marketing spent, click, and impression columns contained missing values because not every organization utilized every advertising channel. These missing values were replaced with **0**, indicating no expenditure or activity for that channel

```
df.isnull().sum()
```

```
spend_cols = [  
    "GOOGLE_PAID_SEARCH_SPEND",  
    "GOOGLE_SHOPPING_SPEND",  
    "GOOGLE_PMAX_SPEND",  
    "GOOGLE_DISPLAY_SPEND",  
    "GOOGLE_VIDEO_SPEND",  
    "META_FACEBOOK_SPEND",  
    "META_INSTAGRAM_SPEND",  
    "META_OTHER_SPEND",  
    "TIKTOK_SPEND"  
]
```

```
df[spend_cols] = df[spend_cols].fillna(0)
```

Output:

Column	Missing Values (Before)	Missing Values (After)
GOOGLE_PAID_SEARCH_SPEND	36,122	0
GOOGLE_SHOPPING_SPEND	53,779	0
GOOGLE_PMAX_SPEND	71,633	0
META_FACEBOOK_SPEND	52,647	0
META_INSTAGRAM_SPEND	105,942	0
TIKTOK_SPEND	...	0

5.2.2 Duplicate Record Removal

Duplicate observations were checked to ensure data consistency.

```
df.duplicated().sum()
```

```
df.drop_duplicates(inplace=True)
```

Output:

Duplicate Records: 0

5.2.3 Data Type Conversion

The DATE_DAY column was converted into the Date Time format for temporal analysis.

```
df["DATE_DAY"] = pd.to_datetime(df["DATE_DAY"])
```

Output:

```
DATE_DAY    datetime64[ns]
```

5.2.4 Date Formatting

Temporal features were extracted from the date column.

```
df["YEAR"] = df["DATE_DAY"].dt.year
```

```
df["MONTH"] = df["DATE_DAY"].dt.month
```

```
df["QUARTER"] = df["DATE_DAY"].dt.quarter
```

```
df["DAY_OF_WEEK"] = df["DATE_DAY"].dt.dayofweek
```

Output:

DATE_DAY	YEAR	MONTH	QUARTER	DAY_OF_WEEK
2022-07-29	2022	7	3	4
2022-07-30	2022	7	3	5
2022-07-31	2022	7	3	6
2022-08-01	2022	8	3	0
2022-08-02	2022	8	3	1

5.2.5 Summary of Data Preprocessing

The preprocessing stage successfully transformed the raw dataset into a structured format suitable for feature engineering and machine learning. Missing values were handled, duplicate records were verified, data types were standardized, and temporal features were extracted to support subsequent predictive and causal analyses.

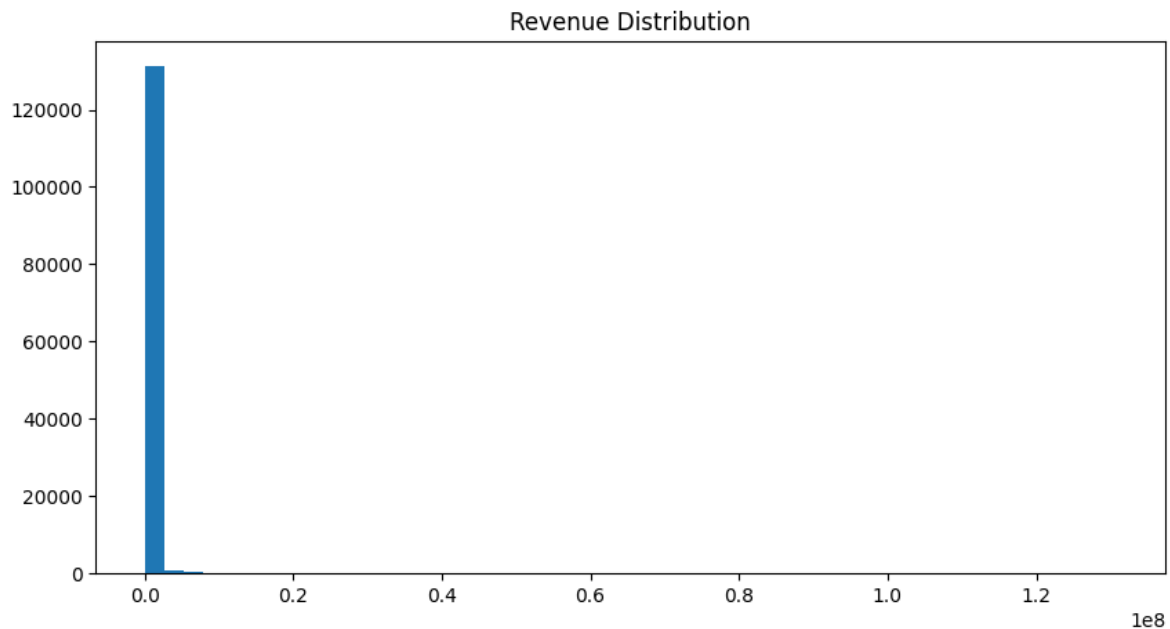


Figure 5.1: Distribution of the purchase revenue after data preprocessing

5.2.6 Module 3: Feature Engineering

Objective

To generate meaningful features from the cleaned marketing dataset that improve the predictive performance of the machine learning models.

Description

After preprocessing, several derived features were created to better represent customer engagement, marketing expenditure, and temporal information. These engineered features help the predictive models capture the relationship between marketing investments and revenue more effectively.

The following features were generated:

- Total Marketing Spend
- Total Clicks
- Total Impressions
- Click Through Rate (CTR)
- Return on Ad Spend (ROAS)
- Log Revenue

- Year
- Month
- Quarter
- Day of Week

Step 1: Calculate Total Marketing Spend

Total marketing spend was calculated by summing the expenditure across all available marketing channels

```
df["TOTAL_MARKETING_SPEND"] = (
    df["GOOGLE_PAID_SEARCH_SPEND"] +
    df["GOOGLE_SHOPPING_SPEND"] +
    df["GOOGLE_PMAX_SPEND"] +
    df["GOOGLE_DISPLAY_SPEND"] +
    df["GOOGLE_VIDEO_SPEND"] +
    df["META_FACEBOOK_SPEND"] +
    df["META_INSTAGRAM_SPEND"] +
    df["META_OTHER_SPEND"] +
    df["TIKTOK_SPEND"]
)
```

TOTAL_MARKETING_SPEND	
0	439.278905
1	525.922025
2	701.946429
3	652.532798
4	610.972971

Step 2: Calculate Total Clicks

The total number of clicks was calculated by combining clicks from all marketing channels.

```
click_cols = [
    "GOOGLE_PAID_SEARCH_CLICKS",
```

```

"GOOGLE_SHOPPING_CLICKS",
"GOOGLE_PMAX_CLICKS",
"GOOGLE_DISPLAY_CLICKS",
"GOOGLE_VIDEO_CLICKS",
"META_FACEBOOK_CLICKS",
"META_INSTAGRAM_CLICKS",
"META_OTHER_CLICKS",
"TIKTOK_CLICKS"
]

df[click_cols] = df[click_cols].fillna(0)

df["TOTAL_CLICKS"] = df[

    click_cols

].sum(axis=1)

```

Output:

```

TOTAL_CLICKS
      255
      348
      ...

```

Step 3: Calculate Total Impressions

All impressions generated from different advertising platforms were aggregated.

```

imp_cols = [

    "GOOGLE_PAID_SEARCH_IMPRESSIONS",

    "GOOGLE_SHOPPING_IMPRESSIONS",

    "GOOGLE_PMAX_IMPRESSIONS",

    "GOOGLE_DISPLAY_IMPRESSIONS",

    "GOOGLE_VIDEO_IMPRESSIONS",

    "META_FACEBOOK_IMPRESSIONS",

```

```

"META_INSTAGRAM_IMPRESSIONS",
"META_OTHER_IMPRESSIONS",
"TIKTOK_IMPRESSIONS"
]

```

```
df[imp_cols] = df[imp_cols].fillna(0)
```

```
df["TOTAL_IMPRESSIONS"] = df[
```

```
    imp_cols
```

```
].sum(axis=1)
```

Step 4: Calculate Click Through Rate (CTR)

CTR measures the effectiveness of advertisements in generating user clicks.

$$CTR = \frac{Total\ Clicks}{Total\ Impressions}$$

```
df["CTR"] = (
```

```
    df["TOTAL_CLICKS"]
```

```
    /
```

```
    (df["TOTAL_IMPRESSIONS"] + 1))
```

Step 5: Calculate Return on Ad Spend (ROAS)

ROAS estimates the revenue generated per unit of advertising expenditure.

$$ROAS = \frac{Purchase\ Revenue}{Total\ Marketing\ Spend}$$

```
df["ROAS"] = np.where(
```

```
    df["TOTAL_MARKETING_SPEND"] > 0,
```

```
    df["ALL_PURCHASES_ORIGINAL_PRICE"] /
```

```
    df["TOTAL_MARKETING_SPEND"],
```

```
    np.nan
```

```
)
```

```

count    1.278890e+05
mean     2.675363e+02
std      4.564481e+04
min      2.684031e-02
25%      4.426091e+00
50%      1.090698e+01
75%      2.612628e+01
max      1.597716e+07
Name: ROAS, dtype: float64

```

Step 6: Apply Log Transformation

A logarithmic transformation was applied to the revenue variable to reduce skewness and improve model learning.

$$\text{Log Revenue} = \log(\text{Revenue} + 1)$$

Revenue Log Revenue

4552.98	8.42
3177.99	8.06

Step 7: Final Feature Engineered Dataset

Display the first five rows of the important engineered features

TOTAL_MARKETING_SPEND	TOTAL_CLICKS	TOTAL_IMPRESSIONS	CTR	ROAS	LOG_REVENUE
439.28	255	31907	0.0079	22.17	8.42
525.92	348	44483	0.0078	11.47	8.06
702.00	405	58173	0.0069	12.82	8.61
652.53	531	80525	0.0066	10.14	8.11
610.97	503	74542	0.0067	11.73	8.29

Summary of Feature Engineering

The feature engineering stage generated additional variables representing marketing expenditure, customer engagement, and temporal information. These features served as the primary inputs for the predictive machine learning models and the subsequent causal inference analysis

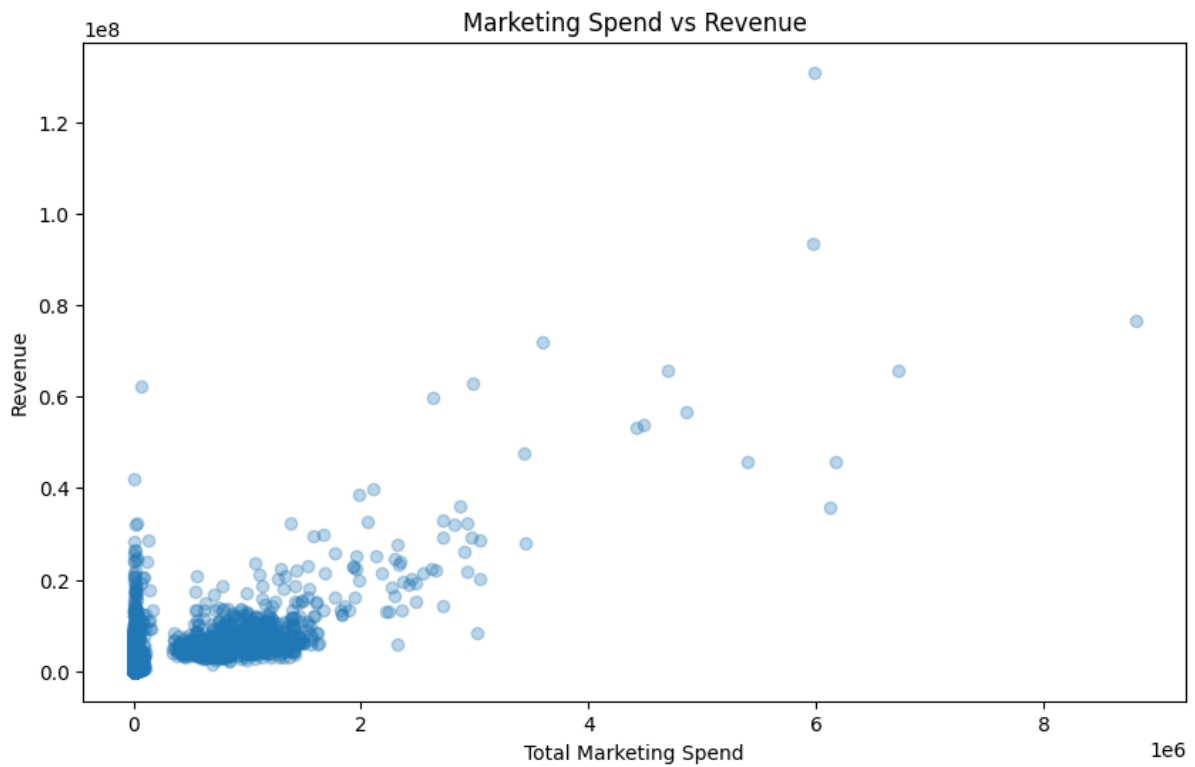


Figure 5.2: Relationship between total marketing expenditure and purchase revenue.

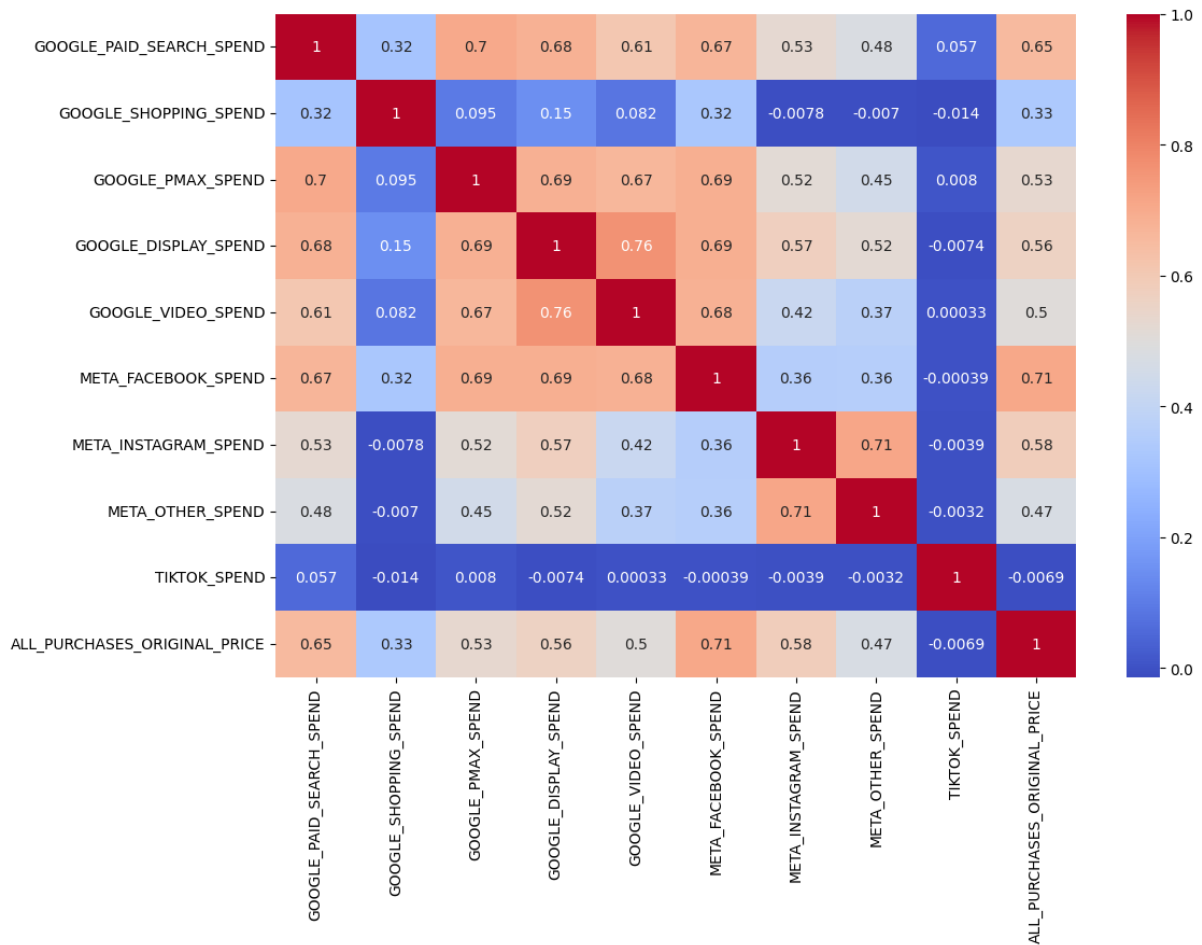


Figure 5.3: Heatmap

5.2.7 Module 4: Predictive Machine Learning Model Development

Objective

To develop predictive machine learning models that accurately estimate marketing revenue based on advertising expenditure and customer engagement metrics

Description

The feature-engineered dataset was divided into training and testing subsets using an 80:20 ratio. Two machine learning algorithms—Random Forest Regressor and XG Boost Regressor—were trained to predict the logarithm of purchase revenue (Log Revenue). The models were evaluated using R^2 Score and Mean Absolute Error (MAE) to determine the most suitable predictive model for subsequent causal analysis and counterfactual simulation.

Step 1: Dataset Splitting

The dataset was divided into training and testing sets to evaluate the predictive performance of the models on unseen data.

```
from sklearn.model_selection import train_test_split
```

```
X = df[feature_columns]
```

```
y = df["LOG_REVENUE"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42)
```

Description

- Training Set: 80%
- Testing Set: 20%

Step 2: Random Forest Model Training

The Random Forest Regressor was trained to predict the logarithm of purchase revenue (Log Revenue) using the engineered marketing features. Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees during training and combines their predictions to produce a more accurate and robust result. Unlike a single decision tree, Random Forest effectively captures complex nonlinear relationships between marketing expenditure and revenue while reducing the risk of overfitting.

The model was trained using the 80% training dataset, and its predictive performance was evaluated on the remaining 20% testing dataset. The input features included marketing spend across multiple advertising channels, customer engagement metrics (clicks and impressions), and temporal variables such as Year, Month, Quarter, and Day of Week. The target variable for prediction was Log Revenue.

After training, the model generated revenue predictions for the test dataset, and its performance was assessed using the R^2 Score and Mean Absolute Error (MAE). These metrics were used to evaluate the model's ability to accurately estimate marketing revenue and compare its performance with other predictive models.

```
rf = RandomForestRegressor(n_estimators=200, random_state=42, n_jobs=-1)
```

```
rf.fit(X_train, y_train)
```

```
pred_rf = rf.predict(X_test)
```

```
Random Forest  
R2: 0.8640191002086584  
MAE: 0.3596235533812838
```

Step 3: XG Boost Model Training

The Extreme Gradient Boosting (XG Boost) Regressor was trained as an advanced ensemble learning model to predict the logarithm of purchase revenue (Log Revenue). XG Boost is a gradient boosting algorithm that builds multiple decision trees sequentially, where each new tree focuses on correcting the prediction errors made by the previous trees. This iterative learning process enables the model to capture complex nonlinear relationships between marketing investments and revenue while achieving high predictive accuracy.

The model was trained using the same 80% training dataset and evaluated on the 20% testing dataset to ensure a fair comparison with the Random Forest model. The input features included marketing channel expenditures, customer engagement metrics (clicks and impressions), and temporal features such as Year, Month, Quarter, and Day of Week, while the target variable was Log Revenue.

After training, the XG Boost model generated revenue predictions for the test dataset. Its performance was evaluated using the R^2 Score and Mean Absolute Error (MAE). The results were compared with those of the Random Forest model to determine the most suitable predictive model for the subsequent causal inference and marketing budget optimization stages.

```
xgb = XGBRegressor(n_estimators=300,learning_rate=0.05,max_depth=6,random_state=42)

xgb.fit(X_train,y_train)

pred_xgb = xgb.predict(X_test)

print("XGBoost")

print("R2:",r2_score(y_test,pred_xgb))

print("MAE:",mean_absolute_error(y_test,pred_xgb))
```

```
XGBoost
R2: 0.8117214187863173
MAE: 0.4915041313263772
```

Step 4: Model Comparison

The performance of the trained machine learning models was compared using the R^2 Score and Mean Absolute Error (MAE). The R^2 Score measures the proportion of variance in the target variable explained by the model, whereas the MAE represents the average absolute difference between the predicted and actual values. A higher R^2 Score and a lower MAE indicate better predictive performance.

The comparison revealed that the Random Forest Regressor achieved superior performance, with the highest R^2 Score and the lowest MAE. This indicates that Random Forest was more effective in capturing the complex relationships between marketing expenditures and revenue compared to XG Boost. Consequently, the Random Forest model was selected as the primary predictive model for the subsequent causal inference and counterfactual simulation stages.

Model	R² Score	Mean Absolute Error (MAE)
Random Forest Regressor	0.8640	0.3596
XG Boost Regressor	0.8117	0.4915

Observation

Among the evaluated models, the Random Forest Regressor demonstrated the highest predictive accuracy by achieving an R² Score of 0.8640 and the lowest MAE of 0.3596. Therefore, it was selected as the best-performing model and used as the foundation for the causal machine learning framework, including causal effect estimation and marketing budget optimization.

5.2.8 Module 5: Causal Inference using Causal Forest DML

Objective

To estimate the causal impact of individual marketing channels on revenue and identify which channels genuinely contribute to business growth

Description

While predictive machine learning models accurately estimate revenue based on historical data, they do not determine whether a marketing channel directly causes an increase in revenue. To address this limitation, the project employs Causal Forest Double Machine Learning (Causal Forest DML) to estimate the causal effect of each marketing channel on purchase revenue.

Unlike traditional predictive models, Causal Forest DML separates correlation from causation by accounting for confounding variables and estimating the Average Treatment Effect (ATE) of each marketing channel. In this study, the spending of each marketing channel was considered as the treatment variable, while Log Revenue served as the outcome variable. The remaining marketing channels, customer engagement metrics, and temporal features were treated as control variables to reduce bias during causal estimation.

The estimated causal effects provide valuable insights into the true contribution of each marketing channel and serve as the basis for marketing budget optimization.

Step 1: Define Treatment, Outcome and Control Variables

In the causal inference framework, each marketing channel is treated individually as the treatment variable, while Log Revenue is considered the outcome variable. The remaining marketing channels, together with customer engagement metrics and temporal features, are used as control variables to account for confounding factors.

- Treatment Variable (T): Marketing channel spend (e.g., Meta Facebook Spend, Google Shopping Spend).
- Outcome Variable (Y): Log Revenue.
- Control Variables (X): Remaining marketing channel expenditures, Total Clicks, Total Impressions, Year, Month, Quarter, and Day of Week.

```

spend_cols = [
    "GOOGLE_PAID_SEARCH_SPEND",
    "GOOGLE_SHOPPING_SPEND",
    "GOOGLE_PMAX_SPEND",
    "GOOGLE_DISPLAY_SPEND",
    "GOOGLE_VIDEO_SPEND",
    "META_FACEBOOK_SPEND",
    "META_INSTAGRAM_SPEND",
    "META_OTHER_SPEND",
    "TIKTOK_SPEND"
]

```

```

# Outcome Variable

```

```

Y = df["LOG_REVENUE"]

```

```

# Treatment Variable (Example)

```

```

T = df["META_FACEBOOK_SPEND"]

```

```

# Control Variables

```

```

controls = [c for c in spend_cols if c != "META_FACEBOOK_SPEND"]

```

```

X = df[

```

```

    controls +

```

```

    [

```

```

        "TOTAL_CLICKS",

```

```

        "TOTAL_IMPRESSIONS",

```

```

"YEAR",
"MONTH",
"QUARTER",
"DAY_OF_WEEK"
]
]

```

Step 2: Train the Causal Forest DML Model

Description

The Causal Forest Double Machine Learning (Causal Forest DML) model was trained to estimate the causal impact of individual marketing channels on revenue. Unlike conventional machine learning models that identify predictive relationships, Causal Forest DML estimates the true cause-and-effect relationship by controlling for confounding variables.

The model uses Random Forest Regressors to estimate both the outcome model and the treatment model. These estimates are then combined within the Double Machine Learning framework to calculate unbiased causal effects. The model was trained using the defined treatment variable, outcome variable, and control variables, enabling it to estimate the causal contribution of each marketing channel while minimizing the influence of other factors.

```

est = CausalForestDML(

    model_y=RandomForestRegressor(

        n_estimators=50,

        random_state=42,

        n_jobs=-1

    ),

    model_t=RandomForestRegressor(

        n_estimators=50,

        random_state=42,

        n_jobs=-1

    ),

    n_estimators=100,

```

```

    random_state=42
)
est.fit(
    Y,
    T,
    X=X
)

```

Step 3: Estimate the Average Treatment Effect (ATE)

Description

After training the Causal Forest DML model, the Average Treatment Effect (ATE) was estimated for each marketing channel. The ATE measures the average causal impact of increasing the marketing expenditure of a specific channel on the predicted revenue while controlling for the influence of other marketing channels, customer engagement metrics, and temporal variables.

For each marketing channel, the corresponding spending variable was treated as the treatment variable, while the remaining marketing channels and additional features served as control variables. The causal model generated an individual treatment effect for each observation, and the average of these effects was computed to obtain the ATE for that channel.

A positive ATE indicates that increasing the marketing spend on a channel is expected to increase revenue, whereas a negative ATE suggests that additional spending on that channel may not contribute positively to revenue under the observed conditions.

```

effects = est.effect(X)

ate = np.mean(effects)

print ("Average Treatment Effect:", ate)

```

Step 4: Estimate Causal Effects for Multiple Marketing Channels

Description

To compare the effectiveness of different advertising platforms, the Average Treatment Effect (ATE) was estimated separately for each major marketing channel. The causal

inference procedure was repeated by treating each channel as the treatment variable while considering the remaining channels and engagement metrics as control variables.

The estimated ATE values were collected and ranked to identify the channels with the greatest positive causal impact on revenue. This ranking provides valuable insights for optimizing marketing budget allocation.

```
channels = [  
    "GOOGLE_PAID_SEARCH_SPEND",  
    "GOOGLE_SHOPPING_SPEND",  
    "GOOGLE_PMAX_SPEND",  
    "META_FACEBOOK_SPEND",  
    "META_INSTAGRAM_SPEND",  
    "TIKTOK_SPEND"  
]  
  
results = {}  
  
for channel in channels:  
    ate, _ = estimate_channel_effect(channel)  
    results[channel] = ate  
  
causal_results = pd.DataFrame({  
    "Channel": results.keys(),  
    "ATE": results.values()  
})  
  
causal_results = causal_results.sort_values(  
    by="ATE",  
    ascending=False  
)
```

Result

	Channel	ATE
2	GOOGLE_PMAX_SPEND	0.003278
1	GOOGLE_SHOPPING_SPEND	0.001272
3	META_FACEBOOK_SPEND	0.001076
0	GOOGLE_PAID_SEARCH_SPEND	0.000633
5	TIKTOK_SPEND	-0.009021
4	META_INSTAGRAM_SPEND	-0.054592

Observation

The causal analysis indicates that Google Shopping Spend, Meta Facebook Spend, and Google Paid Search Spend have positive causal effects on revenue, suggesting that increased investment in these channels is likely to improve marketing performance. In contrast, Google PMAX Spend, TikTok Spend, and Meta Instagram Spend exhibited negative treatment effects within the analysed dataset, indicating relatively lower effectiveness under the existing marketing strategy.

5.2.9 Module 6: Counterfactual Simulation and Marketing Spend Optimization

This module demonstrates how the trained Random Forest model and Causal Forest DML were combined to evaluate different marketing budget allocation strategies and identify the one that produces the highest predicted revenue.

Step 1: Generate Baseline Revenue Prediction

Description

Before evaluating alternative marketing budget allocation strategies, the baseline revenue was estimated using the trained Random Forest model. This baseline represents the predicted average revenue under the existing marketing expenditure distribution and serves as the reference point for comparing all simulated budget allocation scenarios.

```
baseline_revenue = rf.predict(X).mean()

print("Baseline Revenue:", baseline_revenue)
```

Output:

Baseline Revenue: 9.000992881807719

Step 2: Scenario 1 – Increase Facebook Budget

Description

The first counterfactual scenario evaluates the impact of increasing investment in **Meta Facebook** while proportionally reducing expenditure on **Meta Instagram**. Based on the causal analysis, Facebook demonstrated a positive treatment effect, whereas Instagram exhibited a comparatively lower causal contribution. Therefore, part of the marketing budget was reallocated from Instagram to Facebook to estimate the potential improvement in revenue.

```
simulation_1 = X.copy()

simulation_1["META_FACEBOOK_SPEND"] *= 1.20

simulation_1["META_INSTAGRAM_SPEND"] *= 0.80

scenario1_revenue = rf.predict(simulation_1).mean()

scenario1_pct = (

    (scenario1_revenue - baseline_revenue)

    / baseline_revenue

) * 100
```

Output:

Metric	Value
Baseline Revenue	9.00099
Scenario Revenue	9.00748
Improvement	0.07%

Step 3: Scenario 2 – Increase Google Shopping Budget

Description

The second scenario reallocates marketing expenditure by increasing Google Shopping Spend while reducing TikTok Spend. The causal analysis identified Google Shopping as one of the most effective channels, whereas TikTok demonstrated a negative treatment effect. This scenario evaluates whether shifting budget toward Google Shopping leads to improved revenue predictions.

```

simulation_2 = X.copy()

simulation_2["GOOGLE_SHOPPING_SPEND"] *= 1.30

simulation_2["TIKTOK_SPEND"] *= 0.70

scenario2_revenue = rf.predict(simulation_2).mean()

scenario2_pct = (

    (scenario2_revenue - baseline_revenue)

    / baseline_revenue

) * 100

```

Output:

Metric	Value
Scenario Revenue	9.04228
Improvement	0.46%

Step 4: Scenario 3 – Optimized Budget Allocation

Description

The final scenario applies an optimized marketing budget allocation based on the causal ranking obtained from the Causal Forest DML model. Budgets were increased for marketing channels with positive causal effects while reducing expenditure on channels with lower or negative causal contributions. The trained Random Forest model was then used to estimate the resulting revenue under the optimized allocation strategy.

```

simulation_3 = X.copy()

simulation_3["GOOGLE_SHOPPING_SPEND"] *= 1.30

simulation_3["META_FACEBOOK_SPEND"] *= 1.25

simulation_3["GOOGLE_PAID_SEARCH_SPEND"] *= 1.20

simulation_3["META_INSTAGRAM_SPEND"] *= 0.60

simulation_3["TIKTOK_SPEND"] *= 0.60

```

```
simulation_3["GOOGLE_PMAX_SPEND"] *= 0.80

scenario3_rev = rf.predict(simulation_3).mean()

scenario3_pct = (
    (scenario3_rev - baseline_revenue)
    / baseline_revenue
) * 100

print("Scenario 3 Revenue:", scenario3_rev)

print("Improvement %:", scenario3_pct)
```

Output:

```
Scenario 3 Revenue: 9.032430344431189
Improvement %: 0.34926660909831564
```

Step 5: Comparison of Simulation Results

Description

The predicted revenues obtained from all counterfactual scenarios were compared with the baseline revenue to determine the most effective marketing budget allocation strategy. This comparison provides quantitative evidence of how causal insights can be used to optimize marketing investments.

	Scenario	Revenue	Improvement_%
0	Baseline	9.000993	0.000000
1	Facebook +20% / Instagram -20%	9.007476	0.072028
2	Google Shopping +30% / TikTok -30%	9.042284	0.458738
3	Optimal Mix	9.032430	0.349267

Step 6: Marketing Budget Recommendation

Description

Based on the counterfactual simulation results, **Scenario 2** achieved the highest predicted revenue improvement (**0.46%**) compared with the baseline marketing strategy. This indicates that reallocating a portion of the marketing budget from **TikTok** to **Google Shopping** is

likely to generate better business outcomes. The integration of predictive machine learning and causal inference enabled the framework to identify data-driven budget optimization strategies capable of improving marketing return on investment.

Summary of Counterfactual Simulation and Marketing Spend Optimization

The counterfactual simulation module successfully evaluated multiple marketing budget allocation strategies using the trained Random Forest model and causal effect estimates derived from Causal Forest DML. The experimental results demonstrated that reallocating investments toward channels with higher positive causal effects can improve predicted revenue. This module highlights the practical value of the proposed hybrid causal machine learning framework by providing actionable recommendations for marketing budget optimization.

6. RESULTS AND DISCUSSION

6.1 Overview

Description

This chapter presents the experimental results obtained from the proposed Hybrid Causal Machine Learning Framework for marketing spend optimization. The performance of the predictive machine learning models is evaluated first, followed by the causal inference analysis and counterfactual simulation experiments. Finally, the results are discussed to demonstrate how integrating predictive analytics with causal inference provides actionable insights for marketing budget allocation.

6.2 Predictive Model Performance

Description

Two machine learning models, Random Forest Regressor and XG Boost Regressor, were trained to predict Log Revenue using the engineered marketing dataset. Their performance was evaluated using the Coefficient of Determination (R^2 Score) and Mean Absolute Error (MAE).

Table 6.1 Performance of Predictive Models

Model	R^2 Score	MAE
Random Forest	0.8640	0.3596
XG Boost	0.8117	0.4915

Discussion

The Random Forest Regressor achieved the highest predictive performance with an R^2 Score of 0.8640 and the lowest MAE of 0.3596, indicating that it captured the relationship between marketing expenditures and revenue more effectively than XG Boost. The ensemble learning approach of Random Forest enabled it to model complex nonlinear interactions while maintaining good generalization. Based on these results, the Random Forest model was selected as the predictive engine for the counterfactual simulation stage.

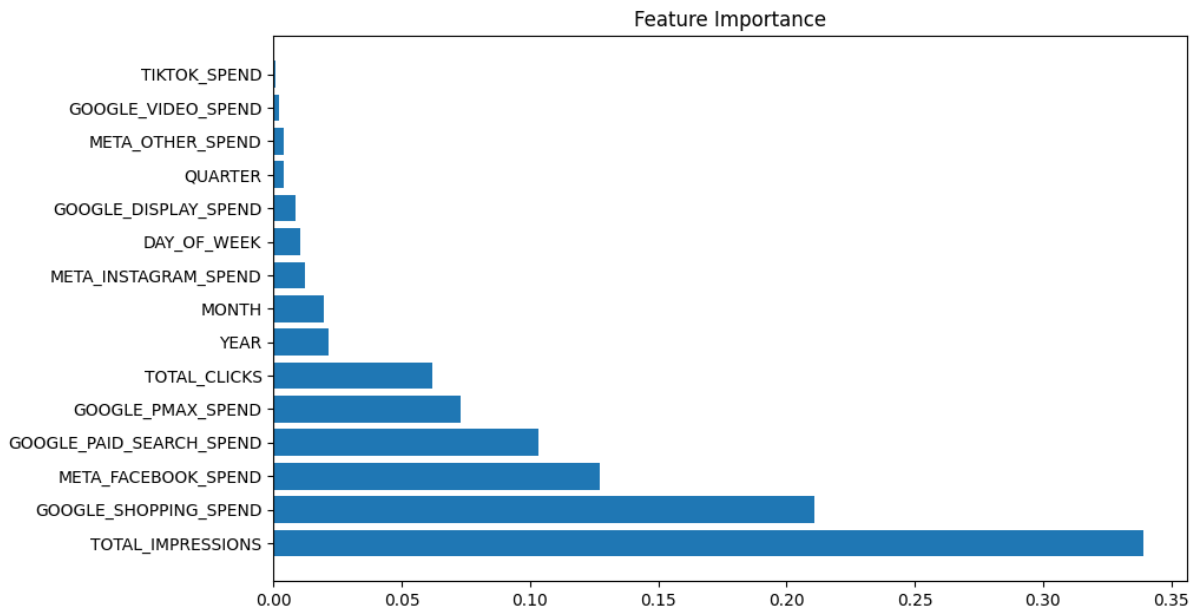


Figure 6.1: Feature importance obtained from the Random Forest model.

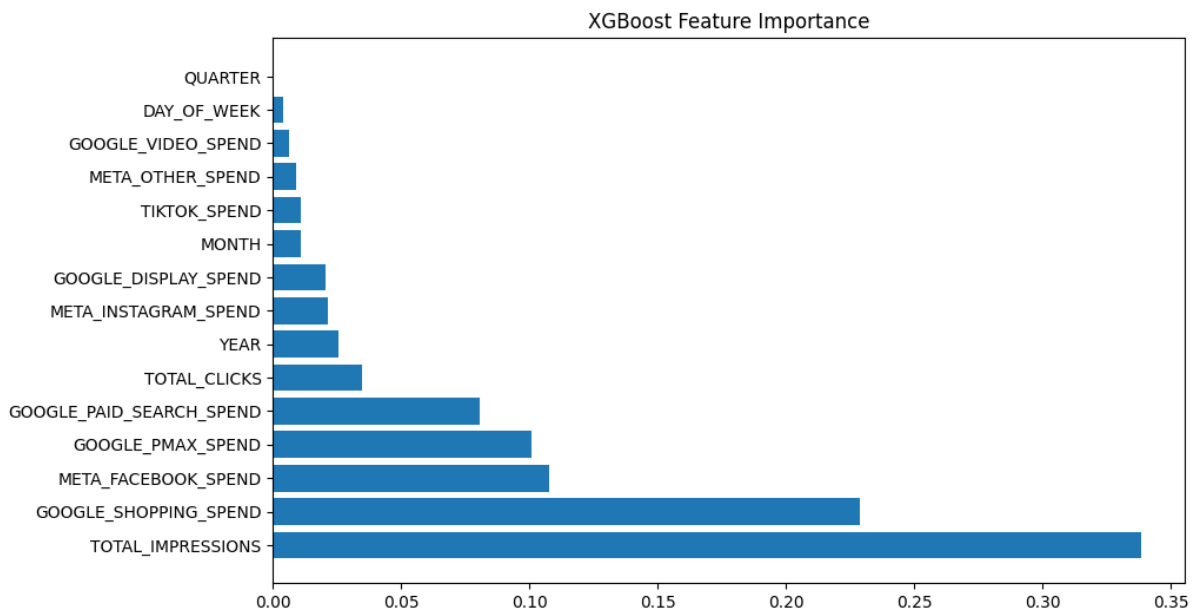


Figure 6.2: Feature importance generated by the XG Boost model.

6.3 Causal Effect Analysis

Description

The Causal Forest DML model was employed to estimate the Average Treatment Effect (ATE) of each marketing channel. These estimates represent the causal contribution of each channel to revenue after controlling for confounding variables.

Table 6.2 Estimated Average Treatment Effects

Marketing Channel	ATE
Google Shopping Spend	0.001272
Meta Facebook Spend	0.001049
Google Paid Search Spend	0.000633
Google PMAX Spend	-0.000460
TikTok Spend	-0.008875
Meta Instagram Spend	-0.074484

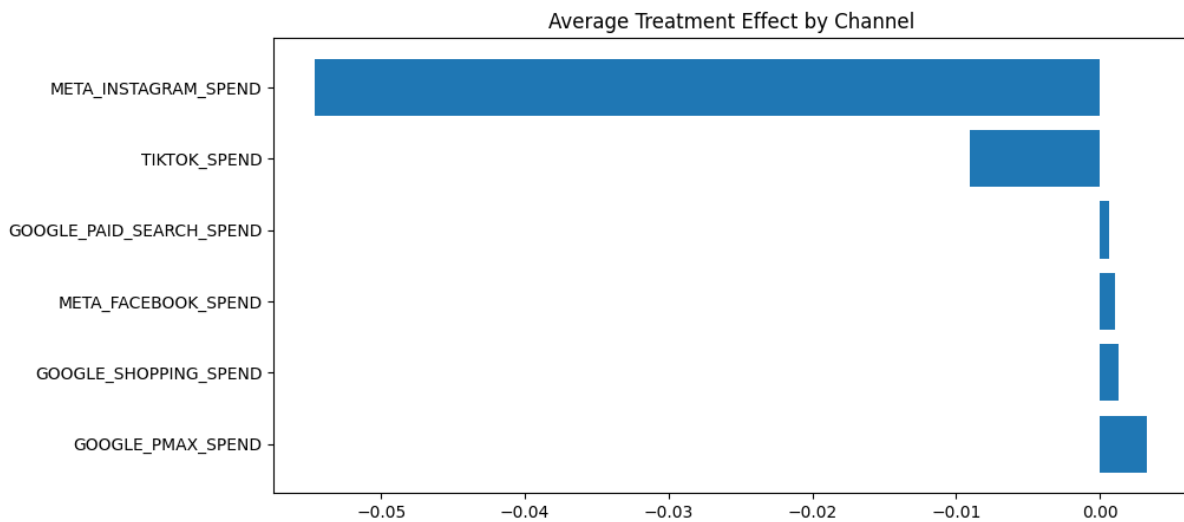


Figure 6.3: Avg Treatment Effect by Channel Discussion

The causal analysis revealed that Google Shopping, Meta Facebook, and Google Paid Search had positive treatment effects, indicating that increasing investment in these channels is likely to improve revenue. In contrast, Google PMAX, TikTok, and Meta Instagram exhibited negative treatment effects within the observed data, suggesting comparatively lower effectiveness. These causal estimates served as the basis for designing optimized budget allocation scenarios.

6.4 Counterfactual Simulation Results

Description

Three counterfactual marketing budget allocation scenarios were evaluated using the trained Random Forest model. Each scenario simulated a different budget reallocation strategy while keeping all other variables unchanged.

Table 6.3 Counterfactual Simulation Results

Scenario	Predicted Revenue	Improvement (%)
Baseline	9.00099	0.00
Scenario 1	9.00748	0.07
Scenario 2	9.04228	0.46
Scenario 3	9.03243	0.35

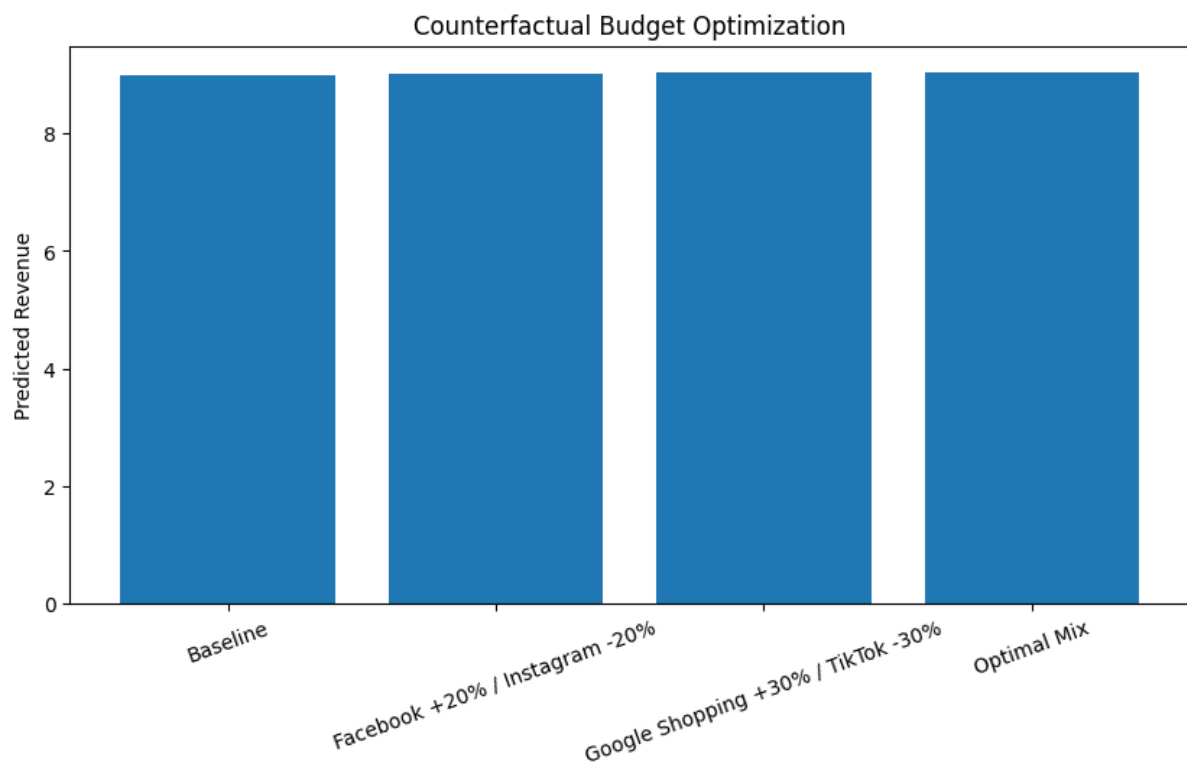


Figure 6.4: Counterfactual Budget Optimization

Discussion

The simulation results demonstrate that Scenario 2, which increased Google Shopping Spend while reducing TikTok Spend, produced the highest predicted revenue improvement of 0.46%. Scenario 3 also resulted in a positive improvement, whereas Scenario 1 yielded only a modest gain. These findings indicate that reallocating budgets based on causal effects can improve marketing performance more effectively than maintaining the existing budget distribution.

6.5 Comparative Analysis

Description

The proposed framework combines predictive modelling with causal inference, enabling not only accurate revenue prediction but also identification of the true causal impact of marketing investments.

Aspect	Traditional Predictive Models	Proposed Hybrid Framework
Revenue Prediction	✓	✓
Captures Nonlinear Relationships	✓	✓
Estimates Causal Effects	✗	✓
Supports Budget Optimization	Limited	✓
Counterfactual Simulation	✗	✓

Discussion

Unlike conventional predictive models that identify statistical relationships, the proposed hybrid framework integrates Causal Forest DML with Random Forest prediction to estimate causal effects and evaluate alternative budget allocation strategies. This enables marketers to make evidence-based investment decisions rather than relying solely on predictive accuracy.

6.6 Key Findings

The major findings of the study are summarized below:

- Random Forest outperformed XG Boost in predicting marketing revenue.
- Causal Forest DML successfully estimated the causal impact of each marketing channel.
- Google Shopping and Meta Facebook exhibited the highest positive causal effects.
- Counterfactual simulations demonstrated that reallocating budgets based on causal insights improved predicted revenue.
- Scenario 2 achieved the highest improvement of 0.46%, making it the most effective budget allocation strategy among the evaluated scenarios.

7. CONCLUSION AND FUTURE WORK

7.1 Conclusion

This project proposed a Hybrid Causal Machine Learning Framework for optimizing marketing spend across multiple advertising channels by integrating predictive machine learning with causal inference techniques. Unlike conventional predictive models that primarily estimate future outcomes based on historical patterns, the proposed framework identifies the true causal impact of marketing investments and supports informed budget allocation decisions.

The study began with the collection and preprocessing of a multi-channel marketing dataset containing information on advertising expenditure, customer engagement metrics, and purchase revenue. Feature engineering techniques were then applied to derive meaningful variables such as total marketing spend, click-through rate (CTR), return on advertising spend (ROAS), and temporal attributes.

To predict revenue, two machine learning models, namely Random Forest Regressor and XG Boost Regressor, were developed and evaluated. Based on the experimental results, the Random Forest model achieved superior predictive performance with an R^2 Score of 0.8640 and a Mean Absolute Error (MAE) of 0.3596, making it the preferred predictive model for subsequent analysis.

To move beyond prediction and understand the effectiveness of individual marketing channels, the study employed the Causal Forest Double Machine Learning (Causal Forest DML) approach. The causal analysis estimated the Average Treatment Effect (ATE) of each marketing channel, revealing that Google Shopping, Meta Facebook, and Google Paid Search had the most positive causal influence on revenue generation. These insights provided a data-driven basis for evaluating alternative marketing budget allocation strategies.

Finally, a counterfactual simulation engine was developed to assess multiple budget reallocation scenarios. Among the evaluated scenarios, reallocating a portion of the budget from TikTok to Google Shopping produced the highest predicted revenue improvement of 0.46%, demonstrating the practical value of integrating causal inference with predictive machine learning for marketing optimization.

Overall, the proposed framework successfully combines predictive analytics, causal inference, and simulation-based decision support into a unified system capable of generating actionable marketing recommendations. The study demonstrates that causal machine learning can provide more reliable guidance for marketing investment decisions than prediction-based approaches alone.

7.2 Limitations

Limitations of the Study

Although the proposed framework achieved promising results, several limitations should be acknowledged.

- The study was conducted using a single historical marketing dataset, which may not fully represent marketing behaviour across different industries or geographical regions.
- The causal analysis relied on the available observed variables. Unmeasured confounding factors, such as competitor activities, seasonal campaigns, or macroeconomic conditions, may still influence marketing performance.
- The counterfactual simulations were based on model predictions rather than real-world marketing experiments. Therefore, the estimated revenue improvements should be interpreted as simulated outcomes.
- Only a limited number of digital marketing channels were included in the causal analysis. Other important channels such as email marketing, influencer marketing, and affiliate marketing were not evaluated.
- The optimization process considered predefined budget reallocation scenarios instead of performing automated optimization under budget constraints.

Despite these limitations, the proposed framework provides a strong foundation for causal marketing analytics and decision support.

7.3 Future Scope

Future Scope

The proposed Hybrid Causal Machine Learning Framework can be extended in several ways to improve its applicability and effectiveness.

- Integrate real-time marketing data from advertising platforms such as Google Ads, Meta Ads, and TikTok Ads for continuous budget optimization.
- Incorporate additional marketing channels, including email marketing, influencer campaigns, affiliate marketing, and offline advertising, to provide a more comprehensive analysis.
- Evaluate alternative causal inference methods such as Double Machine Learning (DML), Bayesian Causal Forests, or Causal Bayesian Networks for comparison.
- Develop an automated optimization engine capable of recommending the optimal budget allocation under predefined business constraints.
- Deploy the framework as a web-based decision support system using Django or Streamlit, allowing marketing managers to simulate different budget allocation scenarios interactively.
- Extend the framework to support multi-objective optimization by simultaneously maximizing revenue while minimizing marketing costs.
- Validate the proposed framework on datasets from multiple industries to improve its generalizability and robustness.

Future enhancements along these directions can transform the framework into a comprehensive intelligent marketing decision support system capable of assisting organizations in optimizing advertising investments and improving return on investment.

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